

MONGODB Practical

1.DATABASE COMMANDS

OneCompiler Upgrade

AI MongoDB Run

script.js

```
1 show dbs
2 use studentDB
3 db
4 db
5 show dbs
6 use studentDB
7 db
8 db
9
```

I/O AI Agent

STDIN
Input for the program (Optional)

Output:
switched to db studentDB
studentDB
studentDB

already on db studentDB
studentDB
studentDB

2.Collection Commands

OneCompiler Up AI MongoDB Run

Save

script.js

```
1 show collections
2 db.createCollection("studengts")
3 db.student.drop()
```

I/O AI Agent

STDIN
Input for the program (Opti

Output:
{ ok: 1 }
true

3. Sample Student Data

script.js

```
1 db.students.insertMany([
2   { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
3   { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
4   { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
5   { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
6   { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
7   { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
8   { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
9   { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
10  { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
11  { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
12 ])
13
```

I/O AI Agent

STDIN
Input for the program (Optional)

Output:
{
 acknowledged: true,
 insertedIds: {
 '0': ObjectId('6a8e58d416688cfb258ce5af'),
 '1': ObjectId('6a8e58d416688cfb258ce5b0'),
 '2': ObjectId('6a8e58d416688cfb258ce5b1'),
 '3': ObjectId('6a8e58d416688cfb258ce5b2'),
 '4': ObjectId('6a8e58d416688cfb258ce5b3'),
 '5': ObjectId('6a8e58d416688cfb258ce5b4'),
 '6': ObjectId('6a8e58d416688cfb258ce5b5'),
 '7': ObjectId('6a8e58d416688cfb258ce5b6'),
 '8': ObjectId('6a8e58d416688cfb258ce5b7'),
 '9': ObjectId('6a8e58d416688cfb258ce5b8')
 }
}

4.Create-insert Commands

```

1 db.students.insertOne({
2   rollNo: 111,
3   name: "Vijay",
4   age: 21,
5   department: "CSE",
6   marks: 86,
7   city: "Hyderabad"
8 })
9
10 db.students.insertMany([
11   { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
12   { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
13 ])
14

```

STDIN

Input for the program (Optional)

Output:

```

{
  acknowledged: true,
  insertedId: ObjectId('6a8e593ecbc477a9e18ce5af')
}

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e593ecbc477a9e18ce5b0'),
    '1': ObjectId('6a8e593ecbc477a9e18ce5b1')
  }
}

```

5.READ – Basic find Commands

```

1 db.students.find()
2 db.students.findOne()
3 db.students.findOne({ rollNo: 101 })
4 db.students.find({ department: "CSE" })
5 db.students.find({ name: "John" })
6 db.students.countDocuments()
7 db.students.countDocuments({ department: "CSE" })
8

```

I/O

AI Agent

STDIN

Input for the program (Optional)

Output:

```

null
null

0
0

```

6.Comparion Operators

```

1 db.students.find({ marks: { $gt: 80 } })
2 db.students.find({ marks: { $gte: 80 } })
3 db.students.find({ marks: { $lt: 80 } })
4 db.students.find({ marks: { $lte: 80 } })
5 db.students.find({ department: { $ne: "CSE" } })
6 db.student.find({marks:$eq:90})
7

```

I/O

AI Agent

1

STDIN

```

db.students.insertMany([
  { name: "Rahul", marks: 85, department: "CSE" },
  { name: "Priya", marks: 92, department: "ECE" },
  { name: "Arjun", marks: 75, department: "CSE" },
  { name: "Sneha", marks: 80, department: "IT" },
  { name: "Kiran", marks: 65, department: "CSE" },
  { name: "Anjali", marks: 90, department: "ECE" }
])

```

Output:

```

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e5b994d53ab868b8ce5af'),
    '1': ObjectId('6a8e5b994d53ab868b8ce5b0'),
    '2': ObjectId('6a8e5b994d53ab868b8ce5b1'),
    '3': ObjectId('6a8e5b994d53ab868b8ce5b2'),
    '4': ObjectId('6a8e5b994d53ab868b8ce5b3'),
    '5': ObjectId('6a8e5b994d53ab868b8ce5b4')
  }
}

```

7.Logical Operators

```
JS script.js x +
1 db.students.find((
2   $and: [
3     { department: "CSE" },
4     { marks: { $gt: 85 } }
5   ]
6 ))
7
8
9 db.students.find((
10   $or: [
11     { department: "CSE" },
12     { department: "IT" }
13   ]
14 ))
15
16
17 db.students.find((
18   marks: { $not: { $gt: 80 } }
19 ))
20
21
22 db.students.find((
23   $nor: [
24     { department: "CSE" },
25     { marks: { $gt: 90 } }
26   ]
27 ))
28

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STDIN
db.students.insertMany([
  { name: "Rahul", department: "CSE", marks: 90 },
  { name: "Priya", department: "CSE", marks: 82 },
  { name: "Arjun", department: "IT", marks: 88 },
  { name: "Sneha", department: "ECE", marks: 95 },
  { name: "Kiran", department: "CSE", marks: 75 },
  { name: "Anjali", department: "IT", marks: 70 }
])

Output:
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e5c67a341e9af288ce5af'),
    '1': ObjectId('6a8e5c67a341e9af288ce5b0'),
    '2': ObjectId('6a8e5c67a341e9af288ce5b1'),
    '3': ObjectId('6a8e5c67a341e9af288ce5b2'),
    '4': ObjectId('6a8e5c67a341e9af288ce5b3'),
    '5': ObjectId('6a8e5c67a341e9af288ce5b4')
  }
}
```

8.IN AND NIN Operators

```
JS script.js x +
1 db.students.find({
2   department: { $in: ["CSE", "IT"] }
3 })
4
5 db.students.find({
6   department: { $nin: ["CSE", "IT"] }
7 })
8

I/O AI Agent
STDIN
db.students.insertMany([
  { name: "Rahul", department: "CSE", marks: 90 },
  { name: "Priya", department: "CSE", marks: 82 },
  { name: "Arjun", department: "IT", marks: 88 },
  { name: "Sneha", department: "ECE", marks: 95 },
  { name: "Kiran", department: "CSE", marks: 75 },
  { name: "Anjali", department: "IT", marks: 70 }
])

Output:
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e5cf859adc1f9df8ce5af'),
    '1': ObjectId('6a8e5cf859adc1f9df8ce5b0'),
    '2': ObjectId('6a8e5cf859adc1f9df8ce5b1'),
    '3': ObjectId('6a8e5cf859adc1f9df8ce5b2'),
    '4': ObjectId('6a8e5cf859adc1f9df8ce5b3'),
    '5': ObjectId('6a8e5cf859adc1f9df8ce5b4')
  }
}
```

9.UPDATE COMMANDS

```
1
2 ∨ db.students.updateOne(
3   { rollNo: 101 },
4   { $set: { marks: 92 } }
5 )
6
7
8 ∨ db.students.updateMany(
9   { department: "CSE" },
10  { $set: { status: "Active" } }
11 )
12
13
14 ∨ db.students.updateOne(
15   { rollNo: 101 },
16   { $inc: { marks: 5 } }
17 )
18
19
20 ∨ db.students.updateOne(
21   { rollNo: 101 },
22   { $inc: { marks: -2 } }
23 )
24
25
26 ∨ db.students.updateMany(
27   {},
28   { $rename: { "city": "location" } }
29 )
30
31
32 ∨ db.students.updateOne(
33   {}
34 )
35

Input for the program ( Optional )

Output:
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

10. DELETE COMMANDS

```
JS scriptjs x +
1 // Delete one
2 db.students.deleteOne({ rollNo: 103 })
3
4 // Delete multiple
5 db.students.deleteMany({ marks: { $lt: 40 } })
6
7 // Delete all documents
8 db.students.deleteMany({})
9
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

```
{ acknowledged: true, deletedCount: 0 }

{ acknowledged: true, deletedCount: 0 }

{ acknowledged: true, deletedCount: 0 }
```

11.SORTING

```
JS scriptjs x +
1 db.students.find().sort({ marks: 1 })
2
3
4 db.students.find().sort({ marks: -1 })
5
6
7 db.students.find().sort({ department: 1, marks: -1 })
8
9
```

I/O AI Agent 1.8s

STDIN

// Descending

Output:

```
===== ASCENDING ORDER =====

[
  {
    _id: ObjectId('6a8e60175b670b3e6d8ce5b4'),
    name: 'Anjali',
    department: 'IT',
    marks: 70
  },
  {
    _id: ObjectId('6a8e60175b670b3e6d8ce5b1'),
    name: 'Arjun',
    department: 'CSE',
    marks: 75
  },
  {
    _id: ObjectId('6a8e60175b670b3e6d8ce5b5'),
    name: 'Ravi',
    department: 'ECE',
    marks: 80
  },
  {
    _id: ObjectId('6a8e60175b670b3e6d8ce5af'),
    name: 'Rahul',
    department: 'CSE',
    marks: 85
  }
]
```

12.LIMIT and SKIP

```
db.students.find().limit(5)

db.students.find().sort({ marks: -1 }).limit(3)

db.students.find().skip(5)

db.students.find().skip(2).limit(3)
```

1.1 LIMIT

toArray()

Output:

```
{
  _id: ObjectId('6a8e60ea9a3e8a70b08ce5af'),
  name: 'Rahul',
  department: 'CSE',
  marks: 85
},
{
  _id: ObjectId('6a8e60ea9a3e8a70b08ce5b0'),
  name: 'Priya',
  department: 'ECE',
  marks: 92
},
{
  _id: ObjectId('6a8e60ea9a3e8a70b08ce5b1'),
  name: 'Arjun',
  department: 'CSE',
  marks: 75
},
{
  _id: ObjectId('6a8e60ea9a3e8a70b08ce5b2'),
  name: 'Sneha',
  department: 'IT',
  marks: 88
},
{
  _id: ObjectId('6a8e60ea9a3e8a70b08ce5b3'),
  name: 'Ravi',
  department: 'ECE',
  marks: 80
}
```

Output:

===== TOP 3 STUDENTS =====

```
[
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b3'),
    name: 'Kiran',
    department: 'CSE',
    marks: 95
  },
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b0'),
    name: 'Priya',
    department: 'ECE',
    marks: 92
  },
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b6'),
    name: 'Meena',
    department: 'CSE',
    marks: 90
  }
]
```

```
[
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b1'),
    name: 'Arjun',
    department: 'CSE',
    marks: 75
  },
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b2'),
    name: 'Sneha',
    department: 'IT',
    marks: 88
  },
  {
    _id: ObjectId('6a8e60ea9a3e8a70b08ce5b3'),
    name: 'Kiran',
    department: 'CSE',
    marks: 95
  }
]
```

13.PROJECTION

Script	Output
<pre>1 db.students.find(2 {}, 3 { _id: 0, name: 1, marks: 1 } 4) 5 6 db.students.find(7 {}, 8 { _id: 0, name: 1, department: 1, marks: 1 } 9) 10 db.students.find(11 {}, 12 { age: 0 } 13)</pre>	<pre>STDIN console.log("HIDE AGE:"); console.log(db.students.find({}, { _id: 0, age: 0 }).toArray());</pre> Output: NAME AND MARKS: <pre>[{ name: 'Rahul', marks: 85 }, { name: 'Priya', marks: 92 }, { name: 'Arjun', marks: 75 }, { name: 'Sneha', marks: 88 }, { name: 'Kiran', marks: 95 }]</pre> NAME, DEPARTMENT AND MARKS: <pre>[{ name: 'Rahul', department: 'CSE', marks: 85 }, { name: 'Priya', department: 'ECE', marks: 92 }, { name: 'Arjun', department: 'CSE', marks: 75 }, { name: 'Sneha', department: 'IT', marks: 88 }, { name: 'Kiran', department: 'CSE', marks: 95 }]</pre>

14.DISTINCT VALUES

```

1 db.students.distinct("department")
2
3
4 db.students.distinct("city")
5
6
7 db.students.distinct(
8   "department",
9   { marks: { $gte: 80 } }
10 )
11

```

```

STDIN
db.students.insertMany([
  { name: "Ravi", department: "CSE", marks: 85, city: "Hyderabad" },
  { name: "Priya", department: "ECE", marks: 72, city: "Vijayawada" },
  { name: "Arjun", department: "CSE", marks: 91, city: "Hyderabad" },
  { name: "Sneha", department: "EEE", marks: 78, city: "Guntur" },
  { name: "Kiran", department: "ECE", marks: 88, city: "Vijayawada" }
])

Output:
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e64b2796cef5a7b8ce5af'),
    '1': ObjectId('6a8e64b2796cef5a7b8ce5b0'),
    '2': ObjectId('6a8e64b2796cef5a7b8ce5b1'),
    '3': ObjectId('6a8e64b2796cef5a7b8ce5b2'),
    '4': ObjectId('6a8e64b2796cef5a7b8ce5b3')
  }
}

```

15. INDEX COMMANDS

```

1 db.students.createIndex({ rollNo: 1 })
2
3 db.students.createIndex({ name: 1 })
4
5
6 db.students.createIndex({ department: 1, marks: -1 })
7
8 db.students.getIndexes()
9
10
11 db.students.dropIndex("name_1")
12
13
14 db.students.dropIndexes()
15

```

```

STDIN
Input for the program ( Optional )

Output:
name_1

department_1_marks_-1

[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { rollNo: 1 }, name: 'rollNo_1' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  {
    v: 2,
    key: { department: 1, marks: -1 },
    name: 'department_1_marks_-1'
  }
]

{ nIndexesWas: 4, ok: 1 }

{
  nIndexesWas: 3,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}

```

16. AGGREGATION-BASIC COMMANDS

```

27 }
28 })
29
30 // Highest marks
31 db.students.aggregate([
32   {
33     $group: {
34       _id: null,
35       highestMarks: { $max: "$marks" }
36     }
37   }
38 ])
39
40 // Lowest marks
41 db.students.aggregate([
42   {
43     $group: {
44       _id: null,
45       lowestMarks: { $min: "$marks" }
46     }
47   }
48 ])
49
50 // Total marks
51 db.students.aggregate([
52   {
53     $group: {
54       _id: null,
55       totalMarks: { $sum: "$marks" }
56     }
57   }
58 ])
59

```

```

STDIN
,
})

Output:
{
  _id: ObjectId('6a8e65c8135ad5ff6b8ce5af'),
  name: 'Ravi',
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
},
{
  _id: ObjectId('6a8e65c8135ad5ff6b8ce5b0'),
  name: 'Priya',
  department: 'ECE',
  marks: 72,
  city: 'Vijayawada'
},
{
  _id: ObjectId('6a8e65c8135ad5ff6b8ce5b1'),
  name: 'Arjun',
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
},
{
  _id: ObjectId('6a8e65c8135ad5ff6b8ce5b2'),
  name: 'Sneha',
  department: 'EEE',
  marks: 78,
  city: 'Guntur'
}

```

17. AGGREGATION WITH MULTIPLE STAGES

js script.js × +

```
1 db.students.aggregate([
2   { $match: { department: "CSE" } },
3   {
4     $group: {
5       _id: "$department",
6       averageMarks: { $avg: "$marks" },
7       totalStudents: { $sum: 1 }
8     }
9   },
10  { $sort: { averageMarks: -1 } }
11 ])
12
13 // Top 3 students using aggregation
14 db.students.aggregate([
15   { $sort: { marks: -1 } },
16   { $limit: 3 },
17   {
18     $project: {
19       _id: 0,
20       rollNo: 1,
21       name: 1,
22       marks: 1
23     }
24   }
25 ])
26
```

I/O AI Agent

STDIN
,
]]

Output:
{
 acknowledged: true,
 insertedIds: {
 '0': ObjectId('6a8e663d4fe75713868ce5af'),
 '1': ObjectId('6a8e663d4fe75713868ce5b0'),
 '2': ObjectId('6a8e663d4fe75713868ce5b1'),
 '3': ObjectId('6a8e663d4fe75713868ce5b2'),
 '4': ObjectId('6a8e663d4fe75713868ce5b3')
 }
}

[{ _id: 'CSE', averageMarks: 88, totalStudents: 2 }]

[
 { rollNo: 103, name: 'Arjun', marks: 91 },
 { rollNo: 105, name: 'Kiran', marks: 88 },
 { rollNo: 101, name: 'Ravi', marks: 85 }
]